

OTEPIC

CROP ROTATION



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TRAINING MANUAL HANDBOOK

CROP ROTATION

The practice of growing different types of crops in the same area across a sequence of growing seasons. The practice reduces the reliance of crops on one set of nutrients, pest and weed pressure, along with the probability of developing resistant pests and weeds.

monocropping, is growing of the same crop in the same place for many years in a row this gradually depletes the soil of certain nutrients and selects for both a highly competitive pest and weed community. Without balancing nutrient use and diversifying pest and weed communities, the productivity of monocultures is highly dependent on external inputs that may be harmful to the soil's fertility.

Conversely, a well-designed crop rotation can reduce the need for synthetic fertilizers and herbicides by better using ecosystem services from a diverse set of crops. Additionally, crop rotations can improve soil structure and organic matter, which reduces erosion and increases farm system resilience.

Crop rotation can also mean changing the type of plant grown in an area each time you plant



ADVANTAGES

1. **Improved Soil Health:** Nutrient Balance: Different crops have varying nutrient requirements. For example, legumes (like beans or peas) fix nitrogen in the soil, enriching it for future crops that may need more nitrogen. Rotating crops helps maintain a balanced nutrient profile and reduces the depletion of specific nutrients.
2. **Prevention of Soil Depletion:** Growing the same crop in the same soil year after year depletes specific nutrients. Crop rotation ensures that the soil's fertility is maintained by diversifying nutrient demands and replenishing the soil with organic matter.
3. **Enhanced Soil Structure:** Rotating deep-rooted and shallow-rooted plants improves soil structure. Deep-rooted crops, like alfalfa, help break up compacted soil layers, while shallow-rooted crops improve soil aeration.
4. **Pest and Disease Management:** Disrupting Pest Life Cycles: Many pests and diseases are crop-specific, so rotating crops can break the life cycle of pests. For instance, planting a different crop after one that attracts a specific pest prevents the pest population from building up in the soil.
5. **Reduced Disease Build-Up:** Planting the same crop repeatedly can lead to an accumulation of soil-borne diseases. Rotating crops reduces the likelihood of pathogens, such as fungi or bacteria, that thrive in the root systems of a particular plant species.
6. **Fewer Pesticides Needed:** By disrupting pest and disease cycles, crop rotation reduces the need for chemical pesticides, making farming more sustainable and environmentally friendly.
7. **Weed Control:** Reduced Weed Pressure: Different crops have different growing habits and can compete with weeds in different ways. For example, rotating crops with different planting and harvest times can suppress weed growth by preventing weeds from adapting to a single crop's growth pattern.
8. **Diverse Weed Control Methods:** Different crops have varying effects on weed growth; for instance, some crops (like rye or oats) can act as allelopathic crops, suppressing the growth of weeds chemically. This can reduce reliance on herbicides.
9. **Increased Crop Yields:** Enhanced Productivity: Crop rotation helps maintain soil fertility, reduces pest pressure, and prevents disease buildup, all of which contribute to higher, more consistent yields. Diversifying crops often leads to better productivity overall compared to monocropping.

10. **Better Use of Land:** By rotating crops, farmers can optimize their land's productivity over time, utilizing different crops that are suited to different conditions and ensuring that the land remains healthy and productive in the long term.
11. **Reduction in Soil Erosion:** Improved Ground Cover: Rotating crops with cover crops (like clover or rye) helps protect the soil from erosion by providing consistent ground cover throughout the year. This helps prevent wind and water erosion, which is especially important on slopes or in areas with heavy rainfall.
12. **Root Structure Stability:** Different crops have different root structures that contribute to soil stability. For example, deep-rooted plants help anchor the soil, reducing the risk of erosion caused by rain or wind.
13. **Enhanced Biodiversity:** Increased Ecological Balance: Crop rotation fosters biodiversity on the farm by encouraging a variety of plant species to grow. This helps support a wider range of beneficial insects, pollinators, and soil organisms, leading to healthier ecosystems.
14. **Habitat for Beneficial Organisms:** The diversity of crops grown in rotation can provide habitats for beneficial organisms like predatory insects, which help control pests naturally.
15. **Economic Benefits:** Reduced Input Costs: Since crop rotation helps manage soil health and pest populations, it can reduce the need for expensive synthetic fertilizers, pesticides, and herbicides. This lowers input costs for farmers.
16. **Risk Diversification:** Growing a range of crops reduces the financial risk associated with crop failure due to pests, diseases, or bad weather. If one crop fails, farmers have others to rely on, spreading financial risk and stabilizing income.
17. **Improved Nitrogen Management:** Nitrogen Fixation: Legumes (such as beans, peas, and clover) fix nitrogen in the soil, enriching the soil for subsequent crops. Rotating legumes with nitrogen-hungry crops (like corn or wheat) reduces the need for synthetic nitrogen fertilizers.
18. **Sustainable Fertilizer Use:** Crop rotation helps reduce the dependency on synthetic fertilizers, promoting more sustainable farming practices and contributing to environmental conservation.
19. **Enhanced Water Efficiency:** Improved Soil Water Holding Capacity: Certain crop rotations, particularly those involving deep-rooted
20. **Control of pests and diseases** which could otherwise thrive due to continuous planting of the same crop

21. **Crops such as beans and other legumes add nutrients to the soil.** This makes the soil stronger and more fertile. Some vegetables take certain nutrients from the soil while growing, this weakens the soil.

By growing the same crop repeatedly in the same area, you cause the soil to be progressively less fertile.

Heavy feeders (weaken the soil)

(strengthen the soil) Maize Lettuce

all types of beans

Cassava Beetroot

Onion Broccoli

Tomato Carrots

almost all other legumes Eggplant etc

Givers

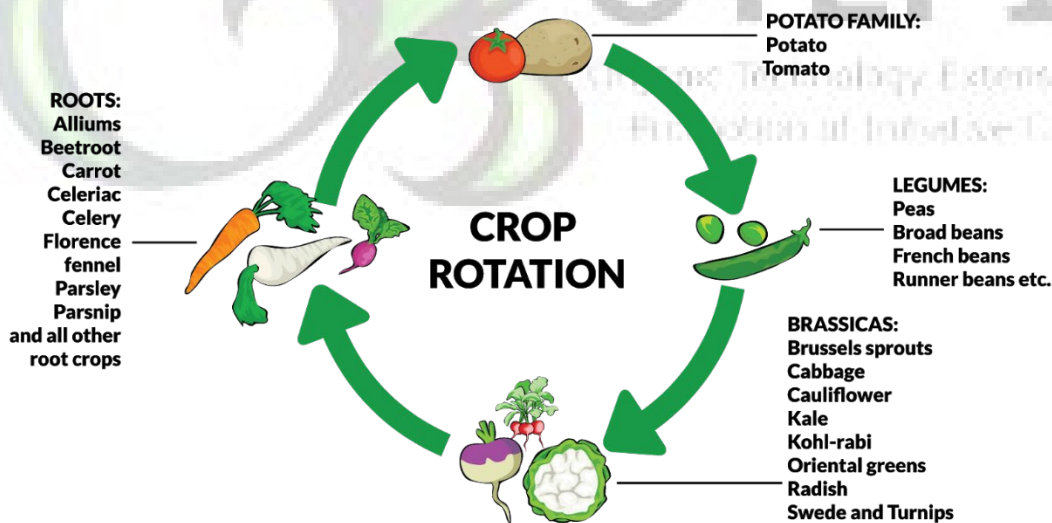
- almost

- groundnuts / peanuts

- peas

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CROP FAMILIES



When you are practicing crop rotation also keep in mind the following crop families as they will help in pest and disease control. The crops in the same family should not be planted in the

following season.

The crop families are as follows:

1. SOLANUMS

- a. Tomatoes.
- b. Egg plants.
- c. Irish potatoes.
- d. Peppers..
- e. Solanum nigrum.

b. Peas.

- c. Ground nuts.
- d. Green and black grams.
- e. Cowpeas.
- f. Soya beans.
- g. Lab beans.
- h. Winged beans.

2. LEGUMES.

- a. Common peas.

i. Other lesser beans.

***NOTE:** forage legumes can be covered under animal production as pastures and fodder crops.*

3. GRAMINACEA.

- a. Maize.
- b. Sorghum.
- c. Sugar cane.
- d. Rice.

- e. Oats.
- f. Finger millet.
- g. Bulrush millet.
- h. Wheat.
- i. Barley

***NOTE:** other grasses can be covered under animal production as pastures and fodder crop section*